

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No - 10	Team No - 8	Project Title - Fitness & Wellness Mobile App for Member Self-Service
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1. Abstract

The *Fitness & Wellness Mobile App for Member Self-Service* is a smart and user-friendly mobile solution designed to simplify fitness management and enhance the overall wellness experience. The application enables members to independently manage their fitness activities, schedules, memberships, and personal wellness information through a single digital platform. It reduces the need for manual assistance while providing convenient access to essential services anytime and anywhere. The app can provide features such as *member registration, workout and fitness tracking, personalized exercise plans, class and trainer booking, membership management, progress monitoring, notifications, and wellness guidance*. By maintaining personalized member profiles and activity records, the system can deliver a more organized and engaging fitness experience. The primary goal of the application is to promote *self-service, convenience, personalization, and healthy lifestyle management*. With an intuitive interface and centralized digital services, the app improves member engagement, saves time, and supports fitness centers in delivering efficient and modern customer service. Overall, the proposed system provides a scalable digital platform that connects members with fitness and wellness services while encouraging consistent and informed participation in their health and fitness journey.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520080061	E.Pranay Reddy	
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3. Frameworks / Platforms / Software Used

- **Frontend:** React.js
- **Backend:** Node.js with Express.js
- **Database:** MySQL (RDBMS)
- **ORM:** Sequelize
- **Authentication:** JWT (JSON Web Token)
- **API:** RESTful APIs using Node.js/Express
- **Database Concepts:** SQL, ER Modelling, Normalisation, Constraints, Joins, Transactions, Views and Stored Logic
- **Security:** JWT-based authentication and role-based access control
- **Communication:** Twilio for SMS/notification services
- **Testing:** Node.js testing tools
- **Development Tools:** Visual Studio Code, Git and GitHub
- **Deployment/Containerisation:** Docker (if required)

Guide Name and Signature (remarks if any) _____